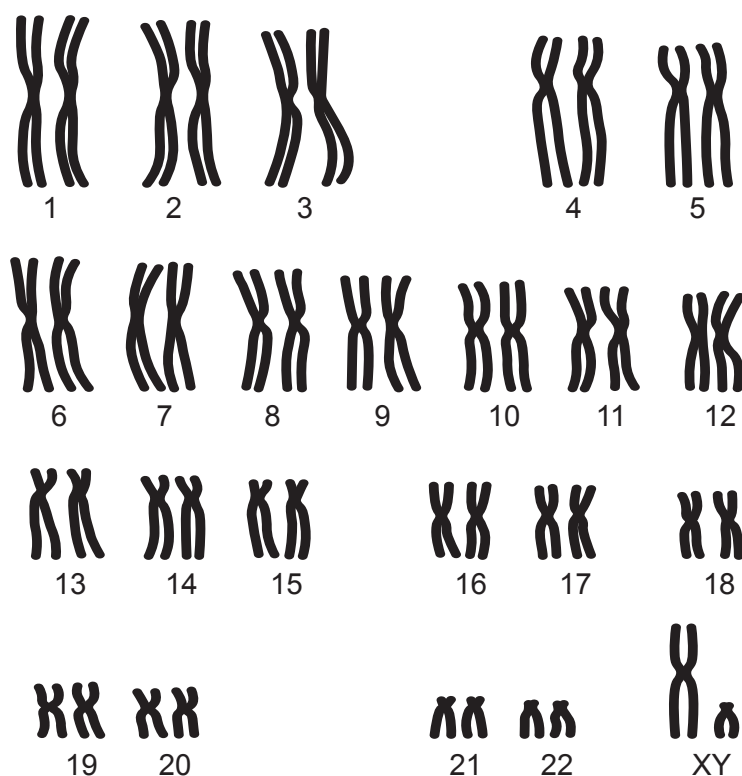


Answer **all** questions

1. **Image 1.1** shows a set of chromosomes from a human male body cell.

Image 1.1

- (a) (i) State the number of chromosomes in a human body cell. [1]
-
- (ii) Draw a **circle** around the sex chromosomes in **Image 1.1**. [1]



- (b) **Images 1.2A and 1.2B** show male and female lions.

Image 1.2A



Male

Image 1.2B



Female

- (i) Complete the Punnett square below to show possible offspring when the male lion (XY) and the female lion (XX) mate. [2]

- (ii) State the expected ratio of male:female lions in the offspring. [1]

..... :

- (iii) State whether it is the male or female lion that determines the sex of the offspring. Explain your answer. [2]

.....

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(c) Lions have 38 chromosomes in their body cells.

(i) State how many chromosomes there are in a **sex cell** of a lion. [1]

.....

(ii) State the scientific name for sex cells. [1]

.....

(iii) Name the type of cell division that produces sex cells. [1]

.....

Examiner
only

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